Test mate economia

sabato 26 ottobre 2024 16:34

Es. 1.

Individuare dominio, min e max assoluti

$f(x) = xe^{3-2x^2}$

$D = \mathbb{R}$

$f'(x) = d(x)\cdot e^{3-2x^2} + x \cdot d(e^{3-2x^2})$

$= e^{3-2x^2} + x \cdot e^{3-2x^2} \cdot d(3-2x^2)$

$= e^{3-2x^2} + xe^{3-2x^2}\cdot(-4x)$

$= e^{3-2x^2}(1-4x^2)$

$f'(x) = 0$

$= e^{3-2x^2}(1-4x^2) = 0$

$e^{3-2x^2} \neq 0$

$-4x^2+1=0$

$-4x^2=-1$

$4x^2=1$

$x^2=\frac{1}{4}$

$x_{1,2}=\pm\sqrt{\frac{1}{4}}$

$x_{1,2}=\pm\frac{1}{2}$

$f''(x) = d(e^{3-2x^2})\cdot(1-4x^2) + d(1-4x^2)\,e^{3-2x^2}$

$= e^{3-2x^2}\cdot(-4x)(1-4x^2) - 8xe^{3-2x^2}$

$= e^{3-2x^2}\cdot(-4x+16x^3) - 8xe^{3-2x^2}$

$= e^{3-2x^2}(-4x+16x^3-8x)$

$= e^{3-2x^2}(+16x^3-12x)$

$= 4xe^{3-2x^2}(+4x^2-3)$

$f''\left(\frac{1}{2}\right) = e^{3-2x^2}(16x^3-12x)$

$= e^{3-2\left(\frac{1}{2}\right)^2}\left(16\left(\frac{1}{2}\right)^3 - 12\left(\frac{1}{2}\right)\right)$

$= e^{3-2\cdot\frac{1}{4}}\cdot\left(16\cdot\frac{1}{8} - 6\right)$

$= e^{3-\frac{1}{2}}(-4)$

$= e^{\frac{5}{2}}\cdot(-4)$

$= -4e^{\frac{5}{2}}$ <0 max (relativo) $x=\frac{1}{2}$

$f''\left(-\frac{1}{2}\right) = e^{\frac{5}{2}}\left(16\cdot\left(-\frac{1}{2}\right)^3 - 12\cdot\left(-\frac{1}{2}\right)\right)$

$= e^{5/2}\left(16\cdot\left(-\frac{1}{8}\right)+6\right)$

$= e^{5/2}(-2+6)$

$= 4e^{5/2}$

$-\frac{1}{2}$ min rel.

Assoluti min e max

$\lim_{x\to+\infty} f(x) = xe^{3-2x^2}$

$= \infty e^{3-2\infty^2}$

$= \infty \cdot \frac{1}{e^{\infty^2}}$

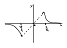
$= 0$

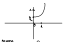
$\lim_{x\to-\infty} -\infty\cdot e^{3-2(-\infty)^2}$

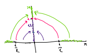
$= -\infty\cdot e^{-2(\infty)^2}$

$= -\infty\cdot\frac{1}{e^{\infty^2}}$

$= 0$







Es. 2

Studio dominio e segno

$f(x) = \log_{\frac{1}{2}}\left(\overbrace{x^2+\frac{3}{4}}^{t}\right)$

$= \log_{\frac{1}{2}}(t) = z$ $\quad t>0$

$= \frac{1}{2}^z = t \neq 0$ $\longrightarrow$ $\lim_{z\to+\infty} \frac{1}{2}^z =$

$\lim_{z\to-\infty} \frac{1}{2}^{(-\infty)} = 0$

$\begin{cases} t \neq 0 \\ t>0 \\ t \not< 0 \end{cases} \longrightarrow \begin{cases} t=1 & f(x)=0 \\ t>1 & f(x)<0 \\ 0<t<1 & f(x)>0 \end{cases}$

$f(x)=0 \to x^2+\frac{3}{4}=1 \to x^2=-\frac{1}{4} \to x_{1,2}=\pm\frac{1}{2}$

$f(x)<0 \to x^2+\frac{3}{4}>1 \to x^2+\frac{3-4}{4}>0 \to x^2-\frac{1}{4}>0 \to x^2>\frac{1}{4} \to \begin{matrix} x_1>\frac{1}{2} \\ x_2<-\frac{1}{2}\end{matrix}$

$f(x)>0 \to 0<\left(x^2+\frac{3}{4}\right)<1$

$-\frac{1}{2}<x<\frac{1}{2}$

$\left(x^2+\frac{3}{4}\right)=0$

$\Delta = b^2-4ac$

$= 0 - 4\cdot1\cdot\frac{3}{4}$

$= -3$ (a parti avrà non tocca asse x)

$x^2+\frac{3}{4}>0$

$x^2>-\frac{3}{4}$ $\{x \in \mathbb{R}\}$

Dominio $= \forall x,\ x \in \mathbb{R}$

Segno $= \begin{cases} f(x)>0 & \left]-\frac{1}{2};\frac{1}{2}\right[\\ f(x)=0 & x=-\frac{1}{2} \ \vee \ x=\frac{1}{2} \\ f(x)<0 & \left]-\infty;-\frac{1}{2}\right[\vee \left]\frac{1}{2};+\infty\right[\end{cases}$

Es. 3

Calcolo Primitiva

$f(x) = \frac{x}{(1+x^2)^3}$

$\int \frac{x}{(1+x^2)^3}\,dx$

$d(u) = d(1+x^2)\,dx$

$= 2x\,dx$

$dx = \frac{du}{2x}$

$f(x) = \frac{\cancel{x}}{u^3}\cdot\frac{du}{2\cancel{x}}$

$= \frac{1}{2u^3}\,du$

$\int f(x) = \int \frac{1}{2u^3}\,du$

$= \int \frac{1}{2}\cdot u^{-3}\,du$

$= \frac{1}{2}\cdot\left(-\frac{1}{2u^2}\right) + c$

$= -\frac{1}{4u^2} + c$

$= \frac{-1}{4(1+x^2)^2} + c$

$= \frac{-1}{4(1+x^4+2x^2)} + c$

$= \frac{-1}{4+4x^4+8x^2} + c$

$= \frac{-1}{4x^4+8x^2+4} + c$

Costo marginale relativo

$f(x) = x^2-4x-5$

determinare x per il costo minimo

$f'(x) = 2x-4$

$=0$

$= 2x = 4$

$x = 2$

controllare se è min o max

$f''(x) = 2$

$f''(2) = 2$ >0 concavità verso l'alto, quindi è un minimo

Es. 5

$\lim_{x\to-\infty} \sqrt{\frac{1+18x^2}{2x^2-18}}$

$= \frac{(1+18x^2)\cdot\frac{1}{2x}}{(2x^2-18)\cdot\frac{1}{2x}}$

$$= \sqrt{\frac{\frac{1}{x^2}+18}{2-\frac{18}{x^2}}}$$

$$= \sqrt{\frac{0+18}{2-0}}$$

$$= \sqrt{\frac{18}{2}}$$

$$= \sqrt{9}$$

$$= 3$$

$$\lim_{x \to 0} \frac{2e^x - 1}{x}$$

$$= \frac{2-1}{0}$$

Impossibile, non esiste

Determinare equazione retta Tangente nel punto $x_0=3$

$$f(x) = x\sqrt{1-x} - 1$$

$$\text{eq. Tangente: } \underbrace{f(x_0)}_{} + \underbrace{f'(x_0)}_{}\overbrace{(x - x_0)}^{}$$

$$= x-(-3) = x+3$$

$$f(-3) = -3\sqrt{1+3} - 1 = -3\sqrt{4} - 1 = -3 \cdot 2 - 1 = -6 - 1 = -7$$

$$f'(x) = \sqrt{1-x} + x \cdot D\left((1-x)^{\frac{1}{2}}\right)$$

$$= \sqrt{1-x} + x \cdot \left(\tfrac{1}{2}\right) \cdot (1-x)^{-\frac{1}{2}} \cdot (-1)$$

$$= \sqrt{1-x} - \frac{x}{2\sqrt{1-x}}$$

$$= \frac{-x+2-2x}{2\sqrt{1-x}}$$

$$= \frac{-3x+2}{2\sqrt{1-x}}$$

$$f'(-3) = \frac{-3(-3)+2}{2\sqrt{1+3}} = \frac{9+2}{2 \cdot 2} = \frac{11}{4}$$

$$t(x_0) = -7 + \frac{11}{4}(x+3)$$

$$= -7 + \frac{11}{4}x + \frac{33}{4}$$

$$= \frac{11}{4}x + \frac{33-28}{4}$$

$$= \frac{11x+5}{4}$$

Es 2

Determinare Grafico e Codominio

$$f(x) = \begin{cases} e^x & x \leq 0 \\ 1-2x & x > 0 \end{cases}$$

$$= \begin{cases} x=0 & f(x)=1 \\ x<0 & f(x) =]0;1[\\ x>0 & f(x) =]-\infty;1[\end{cases} \quad x \leq 0 \quad]0;1]$$

$$f(0) = e^0 = 1$$

$$f(-1) = e^{-1} = \frac{1}{e}$$

$$\lim_{x \to +\infty} 1-2x = 1-\infty = -\infty$$

$$\lim_{x \to 0^+} 1-2x = 1-2\cdot 0^+ = 1$$

$$\lim_{x \to -\infty} e^x = e^{-\infty} = \frac{1}{e^{\infty}} = 0$$

$$\lim_{x \to 0^-} e^x = e^0 = \frac{1}{e^0} = 1$$

$$f(x) =]-\infty;1[\cup]0;1]$$

$$=]-\infty;1]$$